

MEHDI MECHKAK

Senior Applied AI & Systems Engineer | RAG, FastAPI, n8n & Industrial Control

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PROFESSIONAL SUMMARY

Applied AI & Systems Engineer with dual expertise in production LLM architectures (RAG, FastAPI microservices, n8n webhook pipelines, pgvector) and heavy industrial control systems (SCADA, PLC, electrical engineering). Specialized in building auditable, evidence-first support automation, natural language tool builders, and automated ETL pipelines with zero ungrounded responses. Bilingual in English (C2) and French (C1/C2).

ACADEMIC EDUCATION

Licence 3 (Bachelor's Degree) — Électronique, Énergie Électrique & Automatique (EEA) 2025 – Present

Université Paris-Saclay, Orsay, France — *Specialization: Informatique & Réseaux*

- Core Focus: Algorithms, C Programming, Network Architecture & Protocols, Signal Processing, Analog/Digital Electronics, Control Systems & Regulation.

Cycle Ingénieur (Engineering Degree / Master's Program) — Électrotechnique 2021 – 2025

École Nationale Polytechnique (ENP), Alger — *Specialization: Industrial Automation & Power Systems*

- Core Focus: Electrical Power Analysis, SCADA/PLC Control Loops, High-Voltage Protection, Wind Energy & Green Hydrogen Integration.

Classes Préparatoires aux Grandes Écoles (Preparatory Cycle) 2021 – 2023

École Nationale Polytechnique (ENP), Alger

- Intensive two-year competitive prep program in Advanced Mathematics, Physics, Engineering Sciences, and Signal Processing.

PRODUCTION AI & SYSTEMS ENGINEERING PROJECTS

Hotel Concierge AI (ConciergeFlow) — AI Systems Engineer Jan 2024 – Jun 2024

Live Demo: <https://flowarchitect-agency.github.io/hotel-concierge-ai/#demo>

- Architected 24/7 conversational AI concierge for Paris luxury boutique hotels providing sub-2.0s responses on web chat and WhatsApp.
- Constructed n8n workflow pipelines integrated with Airbnb APIs for live room lookups and spa upsell recommendations.
- Built automated regression test harness in Python verifying response times under 2 seconds and 100% grounded inventory answers.

Polylingual AI Support Agent — Applied AI Engineer & Architect Mar 2024 – Aug 2024

Live Demo: <https://d15fb7gio4xmdn.cloudfront.net>

- Designed evidence-first French/English support automation platform using dual hybrid lexical + pgvector semantic retrieval.
- Implemented 78% confidence gating mechanism for human escalations, achieving an **88% auto-resolution rate**.
- Provisioned AWS infrastructure via Terraform (ECS Fargate, RDS PostgreSQL, CloudFront) with Prometheus/Grafana telemetry.

AI Internal Tool Builder (Nexus Control Room) — Full-Stack Engineer & AI Architect Jun 2024 – Nov 2024

GitHub: <https://github.com/FlowArchitect-Agency/ai-internal-tool-builder>

- Built natural language workflow node generator converting EN/FR prompts into visual node maps in <2.1 seconds.
- Enforced atomic server-side state machine approval gates (PENDING, APPROVED, RUNNING) in Next.js 14, Prisma 7, and PostgreSQL.

Automated BI Pipeline with AI Enrichment — Data Engineer & AI Specialist Aug 2024 – Present

GitHub: <https://github.com/FlowArchitect-Agency/automated-bi-pipeline>

- Orchestrated daily Apache Airflow ETL DAG processing raw data across 5 e-commerce channels (€4.97M+ revenue, 20,415 orders).
- Enriched support tickets and reviews using LLM sentiment classification into PostgreSQL data mart powering Streamlit BI dashboards & WeasyPrint PDF reports.

INDUSTRIAL CONTROL & ELECTRICAL ENGINEERING EXPERIENCE

Graduate Engineering Intern — Power Systems & Protection
SONATRACH, Equinor, ENI (Oil & Gas) — Algeria

2024 – 2025

- Conducted detailed analysis of Gas Turbine Generators (GTG) and high-voltage protection systems for 10 kV motors using Megger testing equipment.

Graduate Engineering Intern — Electrical & Instrumentation
SONATRACH — CEPESA (Oil & Gas) — Algeria

2023 – 2024

- Maintained UPS systems, transformers, and rotating machinery. Monitored and optimized Distributed Control Systems (DCS) and PLC logic.

TECHNICAL SKILLS & LANGUAGES

AI & Backend: Python, FastAPI, pgvector, RAG, Next.js 14, TypeScript, n8n, Node.js, SQL, PostgreSQL, Apache Airflow, Docker, Terraform, AWS.

Engineering & Code: C, C++, Java, MATLAB, Signal Processing, SCADA, PLC (Siemens/Schneider), DCS, HMI, Modbus, OPC-UA, LTSpice, AutoCAD.

Languages & Certs: English (C2 Bilingual / EF SET Certified), French (C1/C2 Fluent), Arabic (C2 Native). Wind Energy & Green Hydrogen Training (Univ. Rostock).